

College Project Management & Review Portal (With Machine Learning)

Overview: A web-based academic platform to manage the complete lifecycle of college projects. Students submit projects, tag preferred guides, receive reviews, and track progress digitally. Machine Learning is integrated to add prediction and intelligence to the system.

Problem Statement: Current project management in colleges is largely manual and fragmented. Project submissions through emails or physical copies lead to poor tracking, delayed reviews, lack of transparency, and repeated project topics.

Proposed Solution: A centralized portal where students submit project details and tag a preferred guide. The guide approves or rejects the request and conducts structured reviews. Admins monitor overall project status and analytics.

User Roles:

- **Student:** Submit project, tag guide, upload reports, view review status.
- **Guide:** Approve/reject requests, review submissions, provide feedback, assign marks.
- **Admin:** Monitor users, deadlines, project progress, and reports.

Key Feature – Student Tags the Guide: During submission, the student selects a preferred guide. The review process starts only after guide approval, ensuring consent, workload balance, and realistic academic workflow.

Machine Learning Features:

- Project delay prediction using Random Forest / Logistic Regression
- Duplicate project topic detection using NLP (TF-IDF + Cosine Similarity)
- Student performance risk analysis based on activity and review data

Technology Stack:

Frontend: React.js | Backend: Node.js / Flask | Database: MongoDB

ML: Scikit-learn, NLP | Authentication: JWT

Conclusion: This project provides a structured and intelligent approach to academic project management by combining full-stack development with machine learning for better monitoring and decision-making.